


# Part I
## Contaminated Land Guidelines

Blacktown City Council

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# 1 Introduction

Pursuant to the provisions of the *Environmental Planning and Assessment Act 1979*, Council has a duty of care, when considering Planning Proposals for rezoning, development applications (DAs) and Complying Development Certificates (CDCs), to consider the possibility of land contamination and the implications it has for any proposed future use of land.

In recognition of its duty of care, Blacktown City will adopt a precautionary approach to its consideration of applications involving contaminated or potentially contaminated land. The objective of this approach is to enable any land contamination issues to be identified and dealt with at an early stage in the planning process. In order for this to occur, Council has developed a set of procedures to be followed for Planning Proposals for rezoning, DAs and CDCs. These procedures allow for a merit-based consideration of land contamination issues, i.e. in considering the implications of contamination Council will have regard for the sensitivity of a proposed land use in addition to any technical standards or requirements published by:

(a) The NSW Office of Environment and Heritage (OEH) (incorporating the Environment Protection Authority (EPA)
(b) The Australian and New Zealand Environment Conservation Council (ANZECC)
(c) The National Health and Medical Research Council (NH&MRC)
(d) The National Environment Protection Council (NEPC)

or any other relevant authority.

The objectives of this Part are:

(a) To ensure that potentially contaminated or contaminated land is adequately assessed to determine the extent of contamination

(b) To ensure that potentially contaminated or contaminated land is appropriately remediated so that it is suitable for its proposed use, and does not result in any unacceptable levels of risk to the environment or human health.

The procedural principles which apply to all development proposals in Blacktown are contained in Section 2.1 of this Part of Blacktown Development Control Plan (DCP) 2015. Specific procedures for Planning Proposals, DAs and CDCs are outlined in Sections 3, 4.1 and 4.2 respectively.

The processes of identifying, evaluating and remediating contaminated land are documented in the *"Managing Land Contamination Planning Guidelines: SEPP 55 Remediation of Land"* (Department of Urban Affairs and Planning and EPA, 1998). Council considers these guidelines to be a mandatory reference for consultants assessing contamination levels and undertaking remediation works. Consultants preparing contamination reports should also have a practical working knowledge of the various OEH and NEPC publications on contaminated land including:

(a) EPA, 1994, *Contaminated Sites: Guidelines for Assessing Service Station Sites*

(b) EPA, 1995, *Contaminated Sites: Sampling Design Guidelines*
(c) EPA, 1995, *Contaminated Sites: Guidelines for the Vertical Mixing of Soil on Former Broad-Acre Agricultural Land*
(d) OEH, 2011, *Guidelines for Consultants Reporting on Contaminated Sites*
(e) NSW Department of Environment and Conservation 2006, *Guidelines for the NSW Site Auditor Scheme (2nd edition).*

***Contaminated land*** means land in, on or under which any substance is present at a concentration above the concentration at which the substance is normally present in, on or under (respectively) land in the same locality, being a presence that presents a risk of harm to human health or any other aspect of the environment (*Environmental Planning & Assessment Act 1979*, as amended).

Contamination can result from a number of past and/or present occurrences, such as:

(a) The controlled or uncontrolled disposal of wastes, including sewage
(b) Accidental leakage
(c) Leakage during plant operation, storage or transportation of raw materials, finished products or wastes
(d) The corrosion of underground tanks
(e) The emission of particulate matter into the atmosphere
(f) The migration of contaminants into a site from neighbouring land, either as vapour, leachate or movement of liquids through the soil
(g) The use of agricultural chemicals.

The Table at Appendix 1 provides a reference to some activities that may cause contamination. The Table should be consulted as part of the initial enquiry process undertaken prior to the lodgement of an application with Council. The list of activities in the Table is a guide only to potentially contaminating sources. It should not be used as an exhaustive list.

Council views contamination as a subset of general pollution and will seek, in its assessment and determination of all applications, to ensure the continued compatibility of all development by minimising the potential for polluting discharges, fugitive emissions and controlled spillages by appropriate site management techniques. It is incumbent upon all developers to design and manage their sites in a manner consistent with this objective. Council does not favour site remediation which requires ongoing monitoring of capped contamination.

This Part of the DCP is based on the provisions of the State Government's *Managing Land Contamination: Planning Guidelines (1998), Managing Land Contamination Guidelines (2008)* and *State Environmental Planning Policy (SEPP) No. 55 - Remediation of Land*. These documents, as well as the *Contaminated Land Management (CLM) Act 1997* and *State Environmental Planning Policy (Exempt and Complying Development Codes) 2008* should be consulted in the preparation and assessment of site contamination reports.

# 2 Necessary information in the decision-making process

There are 4 main stages in the investigation of contaminated or potentially contaminated land. At each stage it is the applicant's responsibility to provide the necessary documentation to Council and to fund the work required to prepare such documents.

## 2.1 Stages of investigation

- **Stage 1 – Preliminary**

  This stage involves the investigation and reporting of the site's history and is typically based on readily available information such as historical records of land use, aerial photographs, consultations with previous occupants and relevant authorities, and a site inspection.

  Where contaminating activities are suspected to have had an impact on the land, some initial sampling and analysis will need to be undertaken by a suitably qualified environmental consultant and in accordance with EPA/OEH guidelines, to confirm and support any conclusions reached from the site history appraisal.

  If there are any concerns regarding the existence of asbestos, Council's policy on asbestos management should be consulted. This document titled *Asbestos Management in Blacktown City 2014* provides specific information as to how to deal with asbestos.

- **Stage 2 – Detailed investigation**

  Should the initial investigations fail to clearly demonstrate that the land is suitable for its proposed use, a more detailed assessment and evaluation is required to be undertaken. This detailed evaluation stage involves formal sampling by a suitably qualified environmental consultant in accordance with the relevant EPA/OEH guidelines and standards. Typically, a site specific work plan is developed during this stage, based on previous investigations.

- **Stage 3 – Site Remediation Action Plan**

  The Remediation Action Plan should demonstrate how the proponent proposes to reduce risks to acceptable levels and achieve the desired clean-up levels. The ultimate goal of site clean-up is to select a management strategy which mitigates threats to, and provides protection for, public health, welfare and the environment, as well as allowing flexibility in the future use of the land.

- **Stage 4 – Validation and monitoring**

  The purpose of validation is to confirm whether the pre-determined clean-up objectives have been attained and whether any further remediation or restrictions on land use are required. Ideally, validation should be conducted by the same consultant that conducted the rest of the site investigation and remediation process. The relevant OEH guidelines should be followed in this process.

  Depending on the extent of contamination and the method of remediation, e.g. containment, there may be a need for continual site monitoring and/or restrictions on the development potential of the land. A proper monitoring program should include a monitoring strategy, the parameters to be monitored, the monitoring locations, the frequency of monitoring and the appropriate reporting requirements. Any restrictions on the land's development potential would normally be imposed by Council at the rezoning or DA/CDC determination stage.

  As a general rule, Council prefers remediation strategies that do not rely on site monitoring or land restrictions. Such strategies may include the removal of contaminants and their off-site disposal or their in situ treatment (e.g. bio-remediation).

## 2.2 Preliminary Investigation Report

A Stage 1 Preliminary Investigation Report (including site history information and documentation of known or potential sources of contamination) is required to be submitted to Council in the following circumstances at the nominated time.

- **Planning Proposal**

  Where land is part of an investigation area under Part 3 of the *Contaminated Land Management Act 1997*, where the land has been used for an activity listed at Appendix 1 or where there are incomplete records regarding the historical use of the land and it is proposed to be rezoned for residential, education, recreational, child care or hospital purposes.

  **Nominated time:** At the time of lodging a Planning Proposal for rezoning with Council.

- **Development Applications and Complying Development Certificates**

  Where a change of land use is proposed and where the subject, or immediately adjacent, land is suspected of contamination because it has historically been used for an activity listed at Appendix 1.

  **Nominated time:** As part of the documentation included with the relevant application to Council or a private accredited certifier.

## 2.3 Subsequent stages

Subsequent stages in the investigation of contaminated land need not apply where:

(a) A Preliminary Investigation Report clearly demonstrates, in the opinion of the consent authority, that contamination of the subject land and its surrounds has not occurred

(b) The subject land has previously been remediated to a standard acceptable to the consent authority for the proposed land use.

## 2.4 Site audit

A site audit is an independent review of any or all of the stages of the site investigation process, conducted in accordance with the *Contaminated Land Management Act 1997*. A site audit can only be undertaken by a site auditor accredited by the Office of Environment and Heritage under the *Contaminated Land Management Act 1997*.

Although *State Environmental Planning Policy No. 55 – Remediation of Land* (SEPP No. 55) does not require a mandatory site audit for remediation work, Council may require a site audit at any stage of the planning process if it:

(a) Believes on reasonable grounds that the information provided by the proponent is incorrect or incomplete

(b) Wishes to verify that the information provided by the proponent adheres to appropriate standards, procedures and guidelines

(c) Does not have the internal resources to conduct its own technical review.

## 2.5 Site Auditor Scheme

The current list of Site Auditors as accredited under Part 4 of the *Contaminated Land Management Act 1997* may be obtained from the OEH via the internet site http://www.epa.nsw.gov.au/clm/auditors.htm or telephone 02 9995 5342.

The Site Auditor will produce a Site Audit Report as well as a Site Audit Statement. The OEH Guidelines for the NSW Site Auditor Scheme outline the statutory information requirements to be included in a Site Audit Statement.

Auditors accredited under the CLM Act must complete the Site Audit Statement at the conclusion of an audit and furnish a copy to the party commissioning the audit. If the audit is a statutory requirement under the CLM Act, auditors must also send a copy to the OEH and Council.

A copy of a standard Site Audit Statement can be obtained from the OEH website under information relating to contaminated sites and the Auditor Scheme: http://www.epa.nsw.gov.au/resources/clm/auditorglines06121.pdf.

## 2.6 Independent review

At any time throughout the assessment process Council may request a separate independent review of work done or conclusions drawn by an applicant's consultant.

An independent review may be carried out by a third party, such as another consultant who is qualified to deal with the type of land contamination in question and who is independent of both the applicant and the applicant's consultant.

The cost of an independent review would normally be borne by an applicant and not Council. It may be required prior to any Council decision being made or as a counter-check of work done following an initial Council decision.

# 3 Procedures for Planning Proposals

SEPP No. 55 requires consideration of contamination issues when rezoning land. If a rezoning allows a change of use that may increase the risk to health or the environment from contamination, then Council must be satisfied that the land is suitable for the proposed use or can be remediated to make it suitable. If remediation is necessary, Council must be satisfied that suitable planning controls are in place to ensure that this occurs or it may need to be satisfied that appropriate remediation has taken place prior to proceeding with the rezoning.

The key requirements of this procedure are:

(a) The submission, by the applicant, of a Stage 1 Preliminary Investigation Report at the time of lodgement of the application with Council, in circumstances identified at Appendix 1 of this Part

(b) Should Council's assessment of the Preliminary Investigation Report indicate the need for further information, or where it has not been clearly demonstrated that land is suitable for its proposed use in its present state, Council will require the submission of a Detailed Investigation Report

(c) Should the Preliminary Investigation Report, and/or any additional information submitted, clearly and unequivocally show that contamination is not present, Council may proceed to consider the preparation of a draft Local Environmental Plan (LEP)

(d) If land is found to be contaminated and an applicant's consultant verifies that remediation to a level suitable for the intended use is practical, Council may proceed to consider the preparation of a draft LEP. If remediation is not practical the applicant may either withdraw the application or reconsider land use options for the subject land or, alternatively, Council will refuse the application

(e) If Council is uncertain about the findings of a Preliminary Investigation Report or a Detailed Investigation Report, it may require an independent review or site audit of work undertaken.

With regard to spot rezoning, in some circumstances it may be necessary for remediation to occur prior to the rezoning becoming effective. In such cases the remediation work will be assessed as a Development Application. The decision as to what level of investigation or remediation is undertaken in relation to spot rezonings rests solely with Council.

For generalised rezonings involving large tracts of land it is often difficult to accurately assess the level of likely contamination. In such cases rezoning studies should attempt to identify potential area "hot spots" and should have regard for possible future sensitive land uses and the suitability of land on which they are proposed.

In some cases contamination will not be an issue, e.g. if there is no change of use or where there is a change to a similar use.

# 4 Procedures for Development Applications and Complying Development Certificates

To determine whether proposed works should be assessed as a Development Application or a Complying Development Certificate on land that is contaminated, first it must be determined what category of contamination applies to the site. State Environmental Planning Policy No 55 – Remediation of Land (SEPP 55) provides more detail regarding the impacts of contamination and remediation on development.

## 4.1 Procedures for development applications

In the majority of cases where Category 1 works apply as defined by the SEPP, a development application must be lodged with Council. Category 1 works typically relate to land that is critical habitat, or otherwise environmentally sensitive, or to remediation works which are separately listed as "designated development" under the *Environmental Planning and Assessment Regulation 2000*. Where remediation is proposed on a site declared to be a "remediation site" under the *Contaminated Land Management Act 1997*, the Minister for Planning and Environment is the consent authority. Applicants and developers should familiarise themselves with the provisions of SEPP 55 in relation to the need for development consent for certain remediation works.

If investigations undertaken as part of the assessment of a DA establish that contamination makes the land unsuitable for the proposed use and that remediation is necessary in order for the development to proceed, the consent authority may, if the remediation requires separate consent under SEPP 55 (i.e. is a Category 1 work) require the applicant to amend the proposal to include a remediation strategy as part of an amended DA.

The key requirements of the procedures for development applications where Category 1 works apply are:

(a) The submission by the applicant of a Preliminary Investigation Report at the time of lodgement of the application with the consent authority, in circumstances identified at Section 2.2 of this Part

(b) Should the assessment of the Preliminary Investigation Report indicate the need for further information, the consent authority may call for additional sampling and/or justification to be submitted, at the applicant's cost

(c) Where it has not been clearly demonstrated that land is suitable for its proposed use in its present state, the consent authority will require the submission of a Detailed Investigation Report (see Section 2.1.2)

(d) Where land is found to be free of contamination or where, in the consent authority's opinion, the extent of contamination does not pose a threat to human health or the environment, or otherwise render the site unsuitable for the proposed development, the consent authority may proceed to determine the application without reference to clean-up standards or remediation requirements

(e) If the land is found to be contaminated, an appropriate Site Remediation Action Plan may be required to be submitted for consideration. Where remediation is not practical, the application will be refused by the consent authority

(f) If the consent authority is uncertain about the findings of a Preliminary Investigation Report or a Detailed Investigation Report, it may require an independent review of work undertaken, as documented in Sections 2.4 to 2.6 of this Part

(g) The consent authority may also require any necessary remediation to be validated by an independent environmental consultant

(h) Depending on the degree of contamination and the remediation strategy proposed (as well as the category of remediation works pursuant to SEPP 55), the consent authority may decide to issue a deferred commencement consent or require the developer or land owner to maintain and monitor the land to ensure the containment of contaminants. As a general rule, remediation which does not require ongoing monitoring is favoured by Council.

Information pertaining to land contamination which has been submitted as part of any relevant planning proposal may be sufficient to satisfy some of the requirements of this Part for DAs.

It is incumbent on an applicant to demonstrate to the consent authority that previous investigations, and/or remediation strategy/works, carried out at any stage in the land development process satisfy the provisions of this Part. Where land has been previously remediated, contamination issues will still need to be considered when land is proposed for redevelopment. The consent authority must be sure that any residual contamination is dealt with to permit the proposed new land use, particularly if clean-up standards have changed or there is on-site encapsulation of contaminated material.

## 4.2 Procedures for Complying Development Certificates

In the majority of cases where Category 2 works apply as defined by the SEPP, a Complying Development Certificate may be lodged either with Council or a private certifier. As part of their role, certifiers are obliged to report potential contamination. Category 2 works apply on land that requires remediation works but is not considered Category 1 works under SEPP 55.

If the remediation works form part of a larger body of works that require a DA, the contamination assessment can be undertaken in isolation as a CDC. Key requirements for Category 2 remediation work that can be assessed as a CDC are as follows:

(a) Council must be notified in writing at least 30 days prior to any remediation works commencing

(b) As part of the CDC, a statement must be issued by a person qualified to assess contaminated sites. This statement must clearly detail that the land is suitable for the proposed use either in its current form (i.e. with no remediation works required) or after remediation has been completed

(c) If the CDC is approved it must contain within the conditions of consent that completion of the remediation works are required in accordance with that statement

(d) Council must be notified within 30 days after the completion of remediation works.

SEPP 55 requires that a Notice of Completion is issued to Council for all remediation work (both Category 1 and 2 works).

Remediation is often carried out in conjunction with other development, to make the land suitable for that development. The SEPP contains the following rules for remediation as ancillary development:

(a) Remediation work may be treated as Category 2 work instead of Category 1 if the only reason it is in Category 1 is that it is ancillary to designated development

(b) Remediation work that meets the criteria for Category 1 work may not be treated as Category 2 just because it is ancillary to development without consent

(c) If Category 1 remediation work is carried out ancillary to development without consent, this does not result in a requirement for consent for that development

(d) If remediation work is designated development under Schedule 3 of the *Environmental Planning and Assessment Regulation 2000* or the provisions of a planning instrument, this does not mean that any associated development is also designated.

# 5 Clean-up Notice remediation

Clause 21 of SEPP 55 indicates that the provisions of the SEPP do not apply to Clean-up Notices.

Any development or activity carried out for the purpose of complying with a Clean-up Notice:

(a) May be carried out without development consent

(b) To the extent that it involves carrying out any remediation work must be carried out in accordance with:

  (i) The Contaminated Land Planning Guidelines

  (ii) The guidelines (if any) in force under the *Contaminated Land Management Act 1997*.

# APPENDIX 1 Some activities that may cause contamination

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| **Agricultural/horticultural activities** | Land heavily treated with persistent chemicals such as arsenic and organochlorine based chemicals (e.g. banana plantations, cotton and sugar cane fields, local orchards and horticultural plantations and market gardens), and organophosphate-based chemicals |
| **Airports** | Hydrocarbons (fuels and organic solvents), oils and heavy metals |
| **Asbestos production and disposal** | Asbestos based waste such as asbestos tailings (usually contains 2% asbestos from asbestos mines) |
| **Battery manufacture and recycling** | May involve the use of metals including lead, manganese, zinc, cadmium, nickel, cobalt, mercury, silver or antimony or sulfuric acid |
| **Breweries/distilleries** | Involving the use and storage of alcohols (ethanol, methanol and esters) |
| **Chemical & petrochemical works** | A variety of contaminants from the manufacture, formulation, recovery and storage of organic and inorganic chemicals, including petrochemicals e.g. tar and bitumen, solvents such as chloroform, trichloroethylene and tetrachloroethylene (e.g. for dry cleaning establishments), fertilisers, pesticides, pharmaceuticals, soaps and detergents, dyestuffs, inks and paints, acids and alkalis, adhesives, explosives, flocculants, foam production, plastics, photography, rubber and solvents |
| **Concrete & brick industry** | In areas with clay deposits, open cuts were created from the extractive activities. These open cuts, which could reach a depth of up to 10 metres, are often progressively filled with waste and, in particular, industrial waste |
| **Defence works** | Possible unexploded ordnances, heavy metals, oils, hydrocarbons and asbestos |
| **Docks and railway land, especially large sidings and depots** | Oils and hydrocarbons (including polyaromatics such as PAHs, diesel fraction), volatile chlorinated hydrocarbons, heavy metals (copper, lead, mercury, chromium and zinc), and paints, arsenic, herbicides and asbestos |
| **Drum reconditioning works** | Chemicals, oils, hydrocarbons, heavy metals, oxides, pesticides, fertilisers, detergents, dyestuffs, ink, paint, acids and alkalis |
| **Dry cleaning establishments** | Involving the use of solvents such as tetrachloroethylene and other chemicals |
| **Electrical manufacturing (transformers)** | Heavy metals, oils, PCBs, PAHs |
| **Gasworks, other local carbonisation plants and ancillary by products works** | Hydrocarbons such as naphtha, especially aromatic hydrocarbons such as PAHs, coal tar derivatives, phenolics, "spent oxide" (iron or calcium oxides containing high concentrations of free sulphur, sulphides, sulphates), cyanides (free and complex) and asbestos |

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| **Heavy engineering installations and engine works, e.g. shipbuilding, car manufacturing, electrical and electronic manufacturing** | Heavy metals, oxides, antifouling paints (copper and tributyl and tin based), lead, asbestos, hydrocarbons, solvents, acids/alkalis, refrigerants (chloro-fluorocarbons, etc.) |
| **Installations involving and processing, use or disposal of radioactive materials** | Cobalt (e.g. gamma sterilisation and medical therapy), strontium, cesium and radium (e.g. research and medical applications), uranium (e.g. research), thorium (e.g. industrial processing of mineral sands), plutonium and tritium isotopes and others |
| **Landfills and other waste disposal, storage and treatment sites including transfer stations** | Putrescible (e.g. food waste), paper, glass, plastics, metals and others generated mainly by householders in municipal landfills, bacteriological contaminants (e.g. infectious waste) and cytotoxic chemicals from clinical waste, pesticides from contaminated drums, ash containing heavy metals from coal fired power stations or other incinerators, aluminium and iron from water treatment residuals, heavy metal based waste from mining activities and liquid waste such as solvents, generated mainly by industry in controlled and/or secured landfills |
| **Marinas** | Anti-fouling paints, copper, tribulyltin (TBT) and materials used in electroplating |
| **Metal/metallurgical industry, e.g. refining or recovery of metals, electroplating, heat treatment and metal finishing** | Phosphates, nitrates, nitrites, sulphates, heavy metals (cadmium, chromium, copper, lead, nickel, zinc, cobalt, etc.) and cyanides (from alkali cleaning), aromatic compounds such as benzene, toluene, xylenes, styrene and chlorinated hydrocarbons (from solvent cleaning and paint removal), polycyclic aromatic hydrocarbons (from colouring and bitumenising agents), PCBs and mineral oils, asbestos and beryllium |
| **Metal mines** | Acids, cyanides and heavy metals such as cadmium, copper and zinc from mine tailings and processing plants, asbestos and sulphates |
| **Mining and extractive industries** | Arsenic, mercury and cyanides, explosives, aluminium, copper chromium, cobalt, lead, manganese, nickel, selenium, zinc and radio-radionuclides |
| **Munition production, testing sites and other explosive industries** | Mercury and lead based compounds, RDX, sulphur, nitrates (organic and inorganic), TNT, detonating devices and others |
| **Oil refineries, petroleum storage and distribution** | Petroleum hydrocarbons and lead from oil refineries, petrol stations, above and underground storage tanks and distribution sites |
| **Paper and printing works** | Ash, hydroxides, peroxides, acids, foaming agents, chelating agents (e.g. DTPA), dyes |
| **Pesticide storage areas, areas where vehicles used for the transport and storage of pesticides are washed and areas where tanks are used to store** | Insecticides, fungicides and herbicides |

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| **pesticides** | |
| **Power stations and electrical works** | Polychlorinated biphenyls (PCBs) from transformers and capacitors, ash, tin, lead, copper, mercury (heavy metals), asbestos and solvents |
| **Printing and photographic shops** | Acids, alkalis, solvents, chromium, ammonium compounds, alcohols, silver, potassium, bromide, etc |
| **Scrap yards** | Heavy metals, residues from drums including chlorinated hydrocarbon solvents. |
| **Service stations** | Heavy metals, hydrocarbons, oils, aromatic compounds |
| **Smelters, foundries, iron and steel works** | Lead, copper, zinc, cadmium, mercury, aluminium, fluorides and acid from smelters and foundry operations, PAHs (Poly Aromatic Hydrocarbons), cyanides, heavy metals and benzene, toluene and xylene (BTX), ethyl benzene, asbestos, naphthalene, tars and ammonium sulphate from steel works |
| **Stock dipping, e.g. activities of cattle tick and sheep dip sites** | Chemicals (mainly tickicides) disposed of in the 1960s and 1970s such as DDT, arsenic, BHC, delnev, carbaryl and ethion, current chemicals in use (e.g. taktic, amitraz, cypermethrin, bendiocarb, deltamethrin, flumethrin, bayticol, diazinon, chlorfenvinfos and barricade 's') |
| **Tanneries and associated trades** | Materials derived from the hides and skins such as grease and dung; chemicals used in the preservation and tanning process such as sulphate, lime, sulphide, hydroxides, chlorides and arsenic based compounds for pretreatment, tannins, sulphites and chromium salts for tanning |
| **Termite/ant control** | Where substantial areas of soil may be contaminated with organochlorine residues (e.g. dieldrin, heptachlor, chlordane) |
| **Timber treatment and preservation works** | Creosote, polycyclic hydrocarbons, polyaromatic hydrocarbons (PAHs), copper, chromium, arsenic, boron and pentachlorophenol (PCP) from industries using or making wood preservatives and other organochlorines |

**NOTE:** This list is a guide only.